Chapter 2 Homework Assignment

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Problem 1: Transformations and Standardization

Let X be a continuous random variable.

1. Prove $Var(aX + b) = a^2Var(X)$.

   Sol.

$$\begin{aligned} Var(aX+b) &= E[(aX+b)^2] - E[(aX+b)]^2 \\ &= E[a^2X^2 + 2abX + b^2] - [aE(X) + b]^2 \\ &= a^2E[X^2] + 2abE[X] + b^2 - (a^2E[X]^2 + 2abE[X] + b^2) \\ &= a^2(E[X^2] - E[X]^2) \\ &= a^2Var(X) \end{aligned}$$

2. Let $Y = \frac{X-\mu}{\sigma}$. Show that $E[Y] = 0$ and $Var(Y) = 1$.

   Sol.

   Let $\mu = E[X],\ \sigma = \sqrt{Var(X)} > 0$, therefore

$$E[Y] = E\left[\frac{X-\mu}{\sigma}\right] = \frac{1}{\sigma}\big(E[X] - \mu\big) = \frac{1}{\sigma}(\mu - \mu) = 0,$$

$$Var(Y) = Var(\frac{1}{\sigma}X - \frac{\mu}{\sigma}) = \left(\frac{1}{\sigma}\right)^2 Var(X-\mu) = \frac{1}{\sigma^2}Var(X) = \frac{1}{\sigma^2}\sigma^2 = 1.$$

3. If $X \sim \text{Exp}(\lambda)$ derive the CDF and verify the memoryless property.

   Sol.

   CDF:

$$F_X(x) = P(X \le x) = \int_0^x \lambda e^{-\lambda t}\,dt = \left[-e^{-\lambda t}\right]_0^x = 1 - e^{-\lambda x}.$$

Thus

$$F_X(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-\lambda x}, & x \geq 0 \end{cases}.$$

Let $s, t \geq 0$. Then

$$P(X > s + t \mid X > s) = \frac{P(X > s + t)}{P(X > s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = P(X > t).$$

Hence the exponential distribution is memoryless.

SPC Link: Why is standardization fundamental in control limits?

Problem 2: Normal Sampling Theory

Let $X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$

1. Derive the distribution of $\bar{X}$.

   Sol.

   $\bar{X} = \frac{\sum_{i=1}^{n} X_i}{n} = \frac{X_1 + X_2 + \cdots + X_n}{n}$, then

   $$E[\bar{X}] = \frac{E[X_1 + X_2 + \cdots + X_n]}{n} = \frac{n\mu}{n} = \mu,$$

   $$Var(\bar{X}) = \frac{Var[X_1 + X_2 + \cdots + X_n]}{n^2} = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}.$$

   Hence

   $$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

2. Show $\dfrac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$.

   Sol.

   Let $S^2 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}$, and we already know the following:

   (1) $\bar{X}$ and S^2 are independent by Basu's theorem.

   (2) If $Z = \frac{X-\mu}{\sigma} \sim N(0, 1)$, then $Z^2 \sim \chi^2_1$.

   (3) If $Y_i \overset{iid}{\sim} \chi^2_1$, then $\sum_{i=1}^{n} Y_i \sim \chi^2_n, \quad i = 1, \ldots, n.$

   (4) If $Q \sim \chi^2_k$, then $M_Q(t) = (1 - 2t)^{-k/2}$ for $t < \frac{1}{2}$.

Compute

$$\begin{aligned}\sum_{i=1}^{n}(X_i-\mu)^2 &= \sum_{i=1}^{n}((X_i-\bar{X})+(\bar{X}-\mu))^2\\ &= \sum_{i=1}^{n}(X_i-\bar{X})^2+n(\bar{X}-\mu)^2.\end{aligned}$$

Then divide by σ^2:

$$\sum_{i=1}^{n}\left(\frac{X_i-\mu}{\sigma}\right)^2=\sum_{i=1}^{n}\left(\frac{X_i-\bar{X}}{\sigma}\right)^2+\left(\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}\right)^2.$$

Let

$$U=\sum_{i=1}^{n}\left(\frac{X_i-\mu}{\sigma}\right)^2,\ V=\sum_{i=1}^{n}\left(\frac{X_i-\bar{X}}{\sigma}\right)^2,\ W=\left(\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}\right)^2.$$

Then $U=V+W$, and we can know

$$U=\sum_{i=1}^{n}Z_i^2\sim\chi^2_n,\ V=\frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i-\bar{X})^2=\frac{(n-1)S^2}{\sigma^2},$$

also $\bar{X}\sim N(\mu,\sigma^2/n)$, hence

$$\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}\sim N(0,1)\Rightarrow W\sim\chi^2_1.$$

By (1) and W is a function of $\bar{X}$ while $V=\frac{(n-1)S^2}{\sigma^2}$ is a function of S^2, we have

$$V \perp\!\!\!\perp W.$$

From $U=V+W$ and independence, we have

$$M_U(t)=M_V(t)\,M_W(t).$$

By (4), we have

$$(1-2t)^{-n/2}=M_U(t)=M_V(t)\,(1-2t)^{-1/2}\Rightarrow M_V(t)=(1-2t)^{-(n-1)/2}.$$

Since $M_V(t)$ is the mgf of χ^2_{n-1}, finally,

$$V=\frac{(n-1)S^2}{\sigma^2}\sim\chi^2_{n-1}.$$

3. Deduce $T = \frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t_{n-1}$.

Sol.

Let $Z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$. From Q2, we know and let $U = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$.

Compute

$$\begin{aligned} T = \frac{\bar{X}-\mu}{S/\sqrt{n}} = \frac{(\bar{X}-\mu)/(\sigma/\sqrt{n})}{(S/\sqrt{n})/(\sigma/\sqrt{n})} = \frac{Z}{S/\sigma} &= \frac{Z}{\sqrt{(S/\sigma)^2}} \\ &= \frac{Z}{\sqrt{(n-1)S^2/(n-1)\sigma^2}} \\ &= \frac{Z}{\sqrt{U/(n-1)}} \end{aligned}$$

Because of the definition of t-distribution：if $Z \sim N(0,1)$、$U \sim \chi^2_\nu$ and $Z \perp\!\!\!\perp U$，then

$$\frac{Z}{\sqrt{U/\nu}} \sim t_\nu.$$

In conclusion,

$$T = \frac{\bar{X}-\mu}{S/\sqrt{n}} = \frac{Z}{\sqrt{U/(n-1)}} \sim t_{n-1}$$

SPC Link: Known vs unknown σ changes limits.

Problem 3: Discrete Structure and Limits

1. Derive $E[X], Var(X)$ for $X \sim \text{Bin}(n,p)$.

Sol.

Let $K_i \overset{iid}{\sim} \text{Bernoulli}(p)$, we compute

$$\begin{aligned} E[K_i] &= 1 \times p + 0 \times (1-p) = p, \\ E[K_i]^2 &= 1^2 \times p + 0^2 \times (1-p) = p, \\ Var(K_i) &= p - p^2 = p(1-p). \end{aligned}$$

Since $X = \sum_{i=1}^n K_i$. Therefore,

$$E[X] = E\left[\sum_{i=1}^n K_i\right] = \sum_{i=1}^n E[K_i] = \sum_{i=1}^n p = np,$$

$$Var(X) = Var\left(\sum_{i=1}^n K_i\right) = \sum_{i=1}^n Var(K_i) = \sum_{i=1}^n p(1-p) = np(1-p).$$

2. Prove the Poisson limit of Binomial.

   Sol.

   Let $X_n \sim \text{Bin}(n,p)$ with $np \to \lambda > 0$ as $n \to \infty$ and $p \to 0$.

$$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

   We compute

$$\binom{n}{k} p^k = \frac{n(n-1)\cdots(n-k+1)}{k!} p^k = \frac{1}{k!}\left(\prod_{j=0}^{k-1} \frac{n-j}{n}\right)(np)^k \to \frac{1}{k!}\lambda^k,\ \text{as}\ \ n \to \infty,$$

$$(1-p)^{n-k} = (1-p)^n (1-p)^{-k} = \left[(1-p)^{1/p}\right]^{np} (1-p)^{-k} \to e^{-\lambda} \cdot 1,\ \text{as}\ \ n \to \infty.$$

   In conclusion,

$$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k} \to \frac{e^{-\lambda}\lambda^k}{k!},\ \text{as}\ \ n \to \infty.$$

3. Give an SPC context favoring Poisson modeling.

   Sol.

   在統計製程管制當中，Poisson 分配最常被用來描述在固定時間或固定區域內，某事件發生的次數

   例如，在半導體製程產線中，工程師檢查每片晶片，並記錄晶片表面上的缺陷數量。

   令 X 為每片晶片的缺陷數量，若缺陷在晶片表上隨機且彼此間獨立，且在固定區域內的缺陷發生數近似於常數，令 λ 為每片晶片的平均缺陷數，則

$$X \sim \text{Poisson}(\lambda) \Rightarrow P(X=x) = \frac{e^{-\lambda}\lambda^x}{x!},\ x = 0, 1, 2, \ldots$$

Problem 4: Mixtures and Overdispersion

1. Show Poisson–Gamma $\Rightarrow$ Negative Binomial.

   Sol.

   Let $X \mid \Lambda = \lambda \sim \text{Pois}(\lambda)$ and $\Lambda \sim \text{Gamma}(\alpha, \beta)$ for $\alpha,\ \beta > 0$, i.e.

$$f_\Lambda(\lambda) = \frac{e^{-\beta\lambda}\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1},\ \lambda > 0.$$

Then

$$f_{X|\Lambda}(x \mid \lambda) = \frac{e^{-\lambda}\lambda^x}{x!},\ x = 0, 1, 2, \ldots$$

So we can compute

$$\begin{aligned}
P(X = x) &= \int_0^\infty f_{X|\Lambda}(x \mid \lambda) f_\Lambda(\lambda)\, d\lambda \\
&= \int_0^\infty \frac{e^{-\lambda}\lambda^x}{x!} \cdot \frac{e^{-\beta\lambda}\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} d\lambda \\
&= \frac{\beta^\alpha}{x!\,\Gamma(\alpha)} \int_0^\infty \lambda^{x+\alpha-1} e^{-(\beta+1)\lambda}\, d\lambda.
\end{aligned}$$

Use $\int_0^\infty \lambda^{t-1} e^{-k\lambda}\, d\lambda = \frac{\Gamma(t)}{k^t}$ with $t = x + \alpha$ and $k = \beta + 1$:

$$\begin{aligned}
P(X = x) &= \frac{\beta^\alpha}{x!\Gamma(\alpha)} \cdot \frac{\Gamma(x+\alpha)}{(\beta+1)^{x+\alpha}} \\
&= \frac{\Gamma(x+\alpha)}{\Gamma(\alpha)x!} \cdot \left(\frac{\beta}{\beta+1}\right)^\alpha \left(\frac{1}{\beta+1}\right)^x.
\end{aligned}$$

Let

$$p = \frac{\beta}{\beta+1},\ 1 - p = \frac{1}{\beta+1},\ p \in [0, 1].$$

Therefore

$$P(X = x) = \frac{\Gamma(x+\alpha)}{\Gamma(\alpha)x!} p^\alpha (1-p)^x,\ x = 0, 1, 2, \ldots$$

It is the pmf of a Negative Binomial distribution:

$$X \sim \text{NegBin}(\alpha, p) \text{ for number of failures } x \text{ before } \alpha \text{ successes}$$

2. Show Beta–Binomial via marginalization.

Sol.

Let

$$X \mid P = p \sim \text{Bin}(n, p),\ P \sim \text{Beta}(\alpha, \beta),\ \alpha, \beta > 0,$$

where

$$f_P(p) = \frac{1}{B(\alpha, \beta)} p^{\alpha-1}(1-p)^{\beta-1},\ 0 < p < 1,$$

and

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}.$$

Also,

$$P_{X|P}(x \mid p) = \binom{n}{x} p^x (1-p)^{n-x},\ x = 0, 1, \ldots, n.$$

Then we compute

$$\begin{aligned}
P(X = x) &= \int_0^1 P_{X|P}(x \mid p) f_P(p) dp \\
&= \int_0^1 \binom{n}{x} p^x (1-p)^{n-x} \cdot \frac{1}{B(\alpha, \beta)} p^{\alpha-1} (1-p)^{\beta-1} dp \\
&= \binom{n}{x} \frac{1}{B(\alpha, \beta)} \int_0^1 p^{x+\alpha-1} (1-p)^{n-x+\beta-1} dp \\
&= \binom{n}{x} \frac{B(x+\alpha,\ n-x+\beta)}{B(\alpha, \beta)} \\
&= \binom{n}{x} \frac{\Gamma(x+\alpha)\Gamma(n-x+\beta)}{\Gamma(n+\alpha+\beta)} \cdot \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)},\ x = 0, 1, \ldots, n.
\end{aligned}$$

It is the Beta–Binomial pmf.

3. Explain why mixtures increase variance.

Sol.

混合分配會使變異數增加，乃至於 Law of total variance ：

$$Var(X) = E\left[Var(X \mid \theta)\right] + Var\left(E[X \mid \theta]\right).$$

其中第一項是「抽樣變異」（Sampling variation），描述在參數 θ 固定時的隨機波動；第二項是「異質性變異」（Heterogeneity effect），描述在參數 θ 隨機時而引入的額外波動。

若不混合，此時 θ 為常數，則 $E[X \mid \theta]$ 為常數，因此

$$Var\left(E[X \mid \theta]\right) = 0 \Rightarrow Var(X) = E\left[Var(X \mid \theta)\right].$$

若混合，此時 θ 為隨機變數且會影響 $E[X \mid \theta]$，則

$$Var\left(E[X \mid \theta]\right) > 0 \Rightarrow Var(X) = E\left[Var(X \mid \theta)\right] + Var\left(E[X \mid \theta]\right) > E[Var(X \mid \theta)].$$

因此混合會增加異質性變異，使得總變異數膨脹（Overdispersion）。

以 Problem 4 的 Poisson−Gamma mixture 為例

已知

$$E[X \mid \Lambda] = Var(X \mid \Lambda) = \Lambda,\ E[\Lambda] = \frac{\alpha}{\beta},\ Var(\Lambda) = \frac{\alpha}{\beta^2}$$

可得

$$\begin{aligned}
Var(X) &= E\left[Var(X \mid \Lambda)\right] + Var\left(E[X \mid \Lambda]\right) \\
&= E[\Lambda] + Var(\Lambda) \\
&= \frac{\alpha}{\beta} + \frac{\alpha}{\beta^2}
\end{aligned}$$

我們知道

$$E[X] = E\big(E[X \mid \Lambda]\big) = E[\Lambda] = \frac{\alpha}{\beta}.$$

因此

$$Var(X) = E[X] + \frac{\alpha}{\beta^2} > E[X].$$

總結來說，對於一個 Poisson 分配，其變異數等於期望值，但經過 Poisso-Gamma 的混和分配後，造成變異數大於期望值，亦即變異數膨脹。

SPC Link: Overdispersion violates p-chart assumptions.